

TOSP sequence

1. Obtain a battery powered gaussmeter and one of the extensible radiation meters.
2. Ask Ops to set the Moeller quads to zero BDL and cycle them
3. Controlled access with gaussmeter and screwdriver in hand (survey along path)
4. Remove polyacrylic plate covering the 4" port on the downstream side of 1H02.
5. Use the radiation meter to check the beam pipe at the exit of 1H02.
6. If cool, reach in with the hall probe and place it in the bore of the quad near the beam pipe. Read the field about midway in angle between a pair of pole tips. Record it.
7. Remove arm and probe and recover port.
8. Remove the cover to the port between 1H03 and 1H03A.
9. Check radiation level of the beam pipe between the magnets.
10. If cool, read fields for each magnet, 3 and 3A, as in step 6.
11. Remove arm and recover port.
12. Exit hall



TEST PLAN WORKSHEET

PROGRAM DEPUTY APPROVAL

PD Signoff: _____ Date: _____
Ops Reviewer Signoff: _____ Date: _____
Expiration Date (max. 90 days from approval): _____
Presentation Required? ☐ yes ☐ no

COMPLETION INFORMATION

Completion Date: _____
Crew Chief Signoff: _____
Comments (partial completion, etc.): _____

NOTE: Information addressing the appropriate content of each of the following sections can be found in Section 2.0 of the Test Plan Instructions.

Test Plan Title: Evaluate focusing at end of Hall A line (attempt 2)

Author(s): Jay Benesch

Date Submitted: 4/4/2000

Revision Number: Rev. 0, (4/4/00 1400)

Brief Purpose of Test

Evaluate focusing strength of quads 1H01 through 1H04A.

Anticipated Benefits

so we don't spend as much time screwing around to get hall A beam profile right

Beam Conditions Required

Complete all of the following tables, entering a value or an X in the appropriate spaces:

Beam Type/Current (enter value)

Beam Type	Beam Current
Beam Off	
Pulsed (std. current = 8 μ A) ^a	X
CW	

a. The standard current for pulsed beam operation is 8 μ A. If your test requires pulsed beam current >8 μ A, then specify the required current and provide a brief explanation next to the specified current.

Beam Energy (select one)

Beam Off	845 MeV (1-pass)	1.645 GeV (2-pass)	2.445 GeV (3-pass)	3.245 GeV (4-pass)	4.045 GeV (5-pass)	Other (specify)
		X		X	X	

Beam Termination Point (select one)

Hall A	Hall B	Hall C	BSY Dump	NE Stub Dump	45 MeV Dump	Other (specify)
X						

Type of Test (select one)

Invasive (disrupts beam delivery)

Non-invasive (does not disrupt beam delivery)

to A only

Time Required

few hours

Preferred Time of Test

whenever

Staff Required to Execute the Test (including contact info)

Ops

Controlled Access Requirements

none

Hardware and/or Software Changes Required

NOTE: If software changes are part of the test plan, include the name of the application, the old revision level, the new revision level, and if applicable, whether or not it is possible to roll back to the old revision level (are there hardware limitations, etc.).

none

Setup Procedure

1. make sure you have a recent save of the hall A line

Test Procedure

1. bring up an xterm. `cd ~mccops/hlapps/optics/fopt2`
- ✓ 2. check quads in A line against ~~halla_combo_3395.snap if at fourth pass.~~
~~halla_combo_4232.snap if fifth.~~ *partial save 1276*
3. at the command line, type `fopt.sh`. This will run a short shell script which calls `fopt` with 4 uA minimum current and uses four correctors near the end of the hall A line. The correctors are given in `testjb_corr.cfg` in the same directory. If you trip the ep ion chambers, reduce the corrector strengths listed in `testjb_corr.cfg` from 500 to, say, 350, (i.e. edit the file) and trying the script again. Add comments to the `fopt` file! e.g. "standard" for this run
4. set the last six quads in the A line: 1H01, 1H04, 1H04A on the quad page and the three on the Moeller page, to zero. Make sure they cycle.
5. run `fopt.sh` again. comment : zero
6. Set MQA1H01 = 13500. Set MQM1H02 = -15767.5 make sure both cycle.
7. run `fopt.sh` again comment: 1H01 vs 1H02

8. Set MQM1H02 = 0 and set MQO1H03 = -14488. Cycle both. Leave 1H01 set at 13500
9. run fopt.sh again. comment : 1H01 vs 1H03 1202
10. Set MQO1H03 = 0 and MQO1H03A = -14488. Cycle both. MQA1H01 no change.
11. run fopt.sh again. comment : 1H01 vs 1H03A 1220
12. restore the end of the hall A line using the save set noted in the Setup section. Cycle the eight magnets used in this test (six quads and 1C20 correctors).

Backout Procedure

1. see step 12 above.

Test Results

$\begin{pmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{pmatrix}$ should be $Me \begin{bmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{bmatrix}$

- 1) y from x_1
- 2) x_2 (really x_1)
- 3) x from y_1
- 4) y_2

1) What I do now.

a) MST

b) LEM $??$

c) machine update script

d) locks

What is appropriate order for setup?

. MT. a) modify calc values
 b) show BEM on accel ~~gene~~ over

Modelserver { ~~don't~~ ^{use} cath's field
 Optim

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To (company name) DEPT OF PHYSICS
UNIVERSITY OF BASEL

Date 29 January 1992

Attention of PROF INGO SICK

Number of pages (including this one) 4

From GARRY FERGUSON

Receiver's fax number 01041 61 267 3784

Ref _____

cc DIETMAR BUDZYLEK

Subject 4 TESLA SPLIT PAIR MAGNET

Don Crabb asked me to contact you with regard to your requirement for a 4 tesla split pair magnet. I have performed a preliminary design of such a magnet to suit one of our standard cryostats (SM4000). The homogeneity of the magnet is 1 part in 100 over a 20m dsv. Some modification of the cryostat is required to suit your 3 inch bore and rectangular split requirements. I have enclosed a field plot of the magnet as designed and some sketches of the cryostat including a cross sectional view in the horizontal plane through the bore and split.

Please contact me on the above number when you have had a chance to have a look at the design. If this design will meet your requirements then I will arrange for a formal quotation to be sent through our German sales office.

Regards

Garry Ferguson

44 865 882855

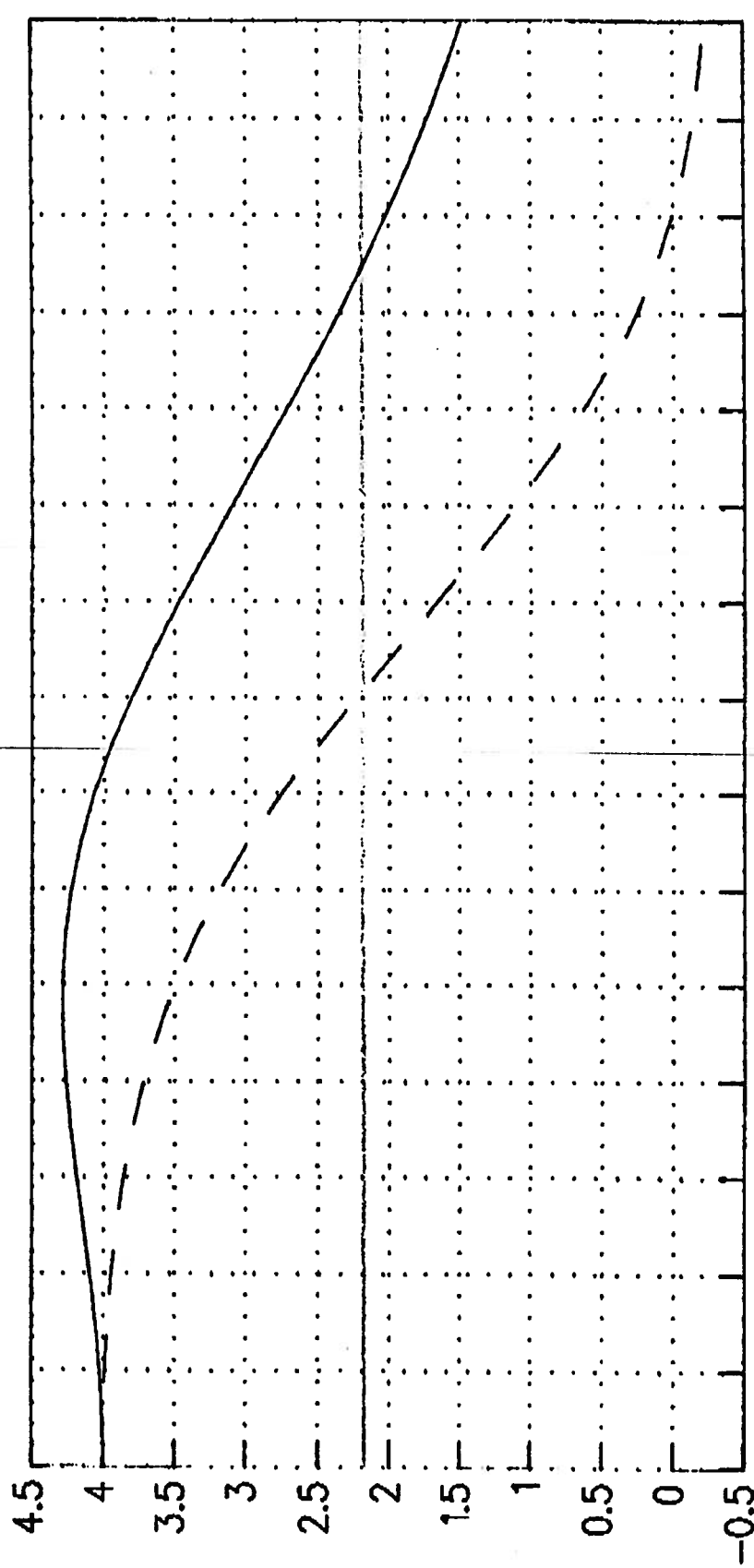
Please call (0865) 882855
if this fax is unclear

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INGO SICK - 4 TESLA SPLIT PAIR

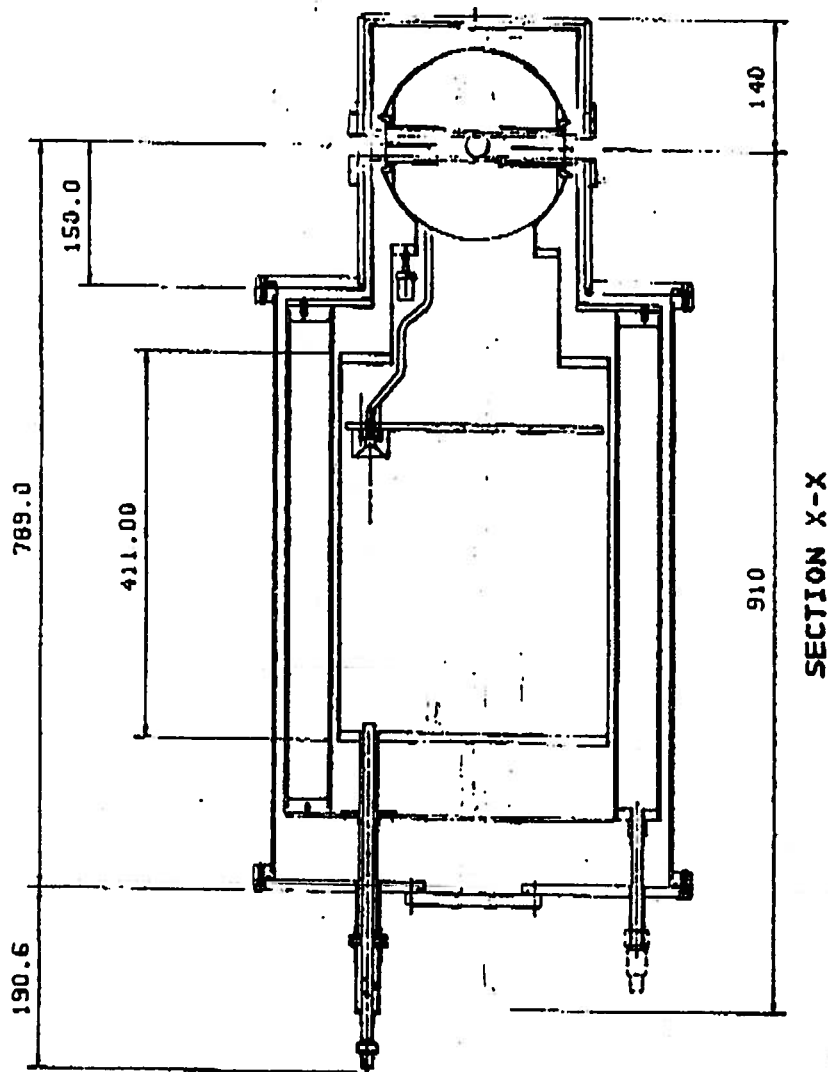
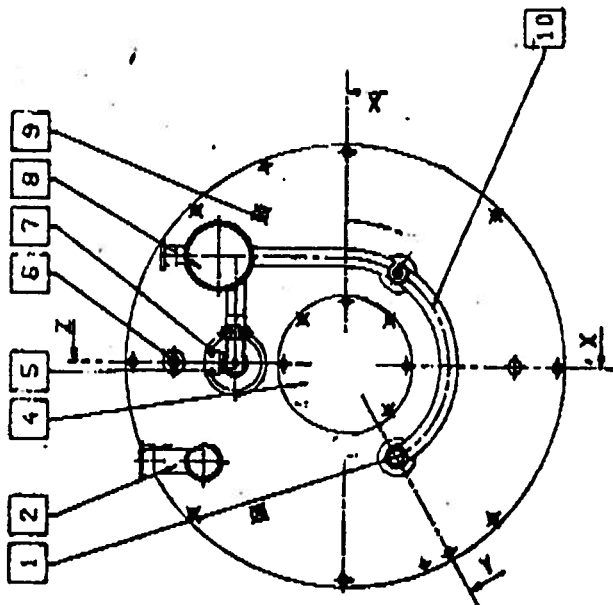
Along Bore Along Split

Field (Tesla)



Displacement (cm)

LEGEND.		
1	VTI NEEDLE VALVE	
2	O.V.C. 0.750 VAC VALVE & SAFETY	MV25
3		
4	SAMPLE SPADE Ø25.0 REF	MU40
5	120000 CO-AX CURRENT LEAD	
6	WIRING FOR NEEDLE VALVE	
7	WIRING FOR H ₂ LEVEL SWITCH & A/B'S	
8	QUENCH VALVE	MU10
9	3 OFF N2 STACKS	
10	HELIUM RECOVERY MANIFOLD	MU10
11		



Massstab 1:2

